Biomedical Materials:
Polymeric biomaterials

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Aim

1. To know physical and chemical properties of polymers
2. To know degredation mechanisms
3. To know some important biopolymers

You have to know

Cathater

Vascular graft

Catheter tubing

Bone graft

Tissue engineering

Tissue scaffold

Drug delivery

Suture

Hydrogel

Some polymers that are used in biomedical research

Polyethylene
PEEK (Polyether ether ketone)
Polyacrylic acid (PAAc)
Polyvinyl Chloride (PVC)
Polyethylene Terephthalate (PET)
Polyurethane (PU)
Polycarbonate (PC)
Poly(methyl methacrylate) (PMMA)
Polyethylene Glycol (PEG)
Poly(glycolic acid) (PGA)
Poly(lactic-co-glycolic acid) (PLGA)
Polydioxanone (PDO)
Polyhydroxyalkanoates (PHA)
Silicone Rubber
Hydrogels: poly(acrylamide)
Chitosan (natural bşopolymer)
Etc.

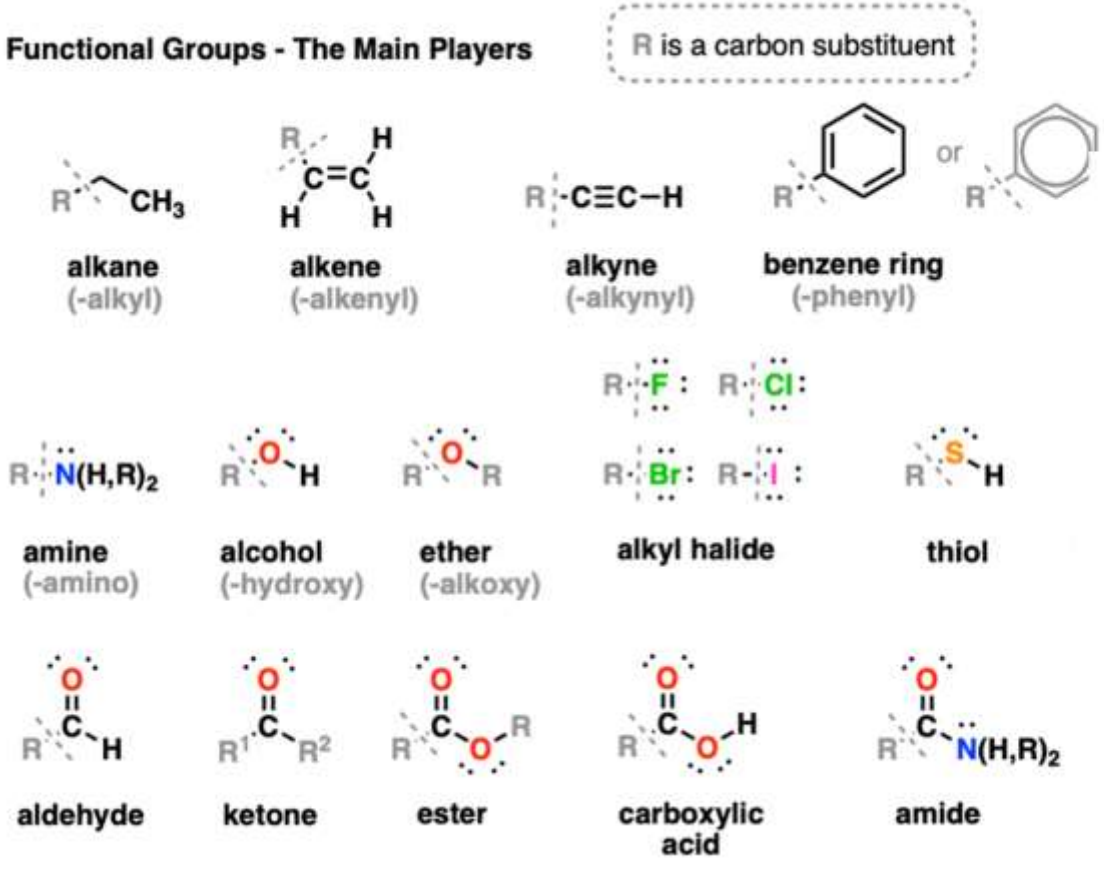
The choice of polymer for a specific application is dictated by multiple factors such as application-specific requirements, including:

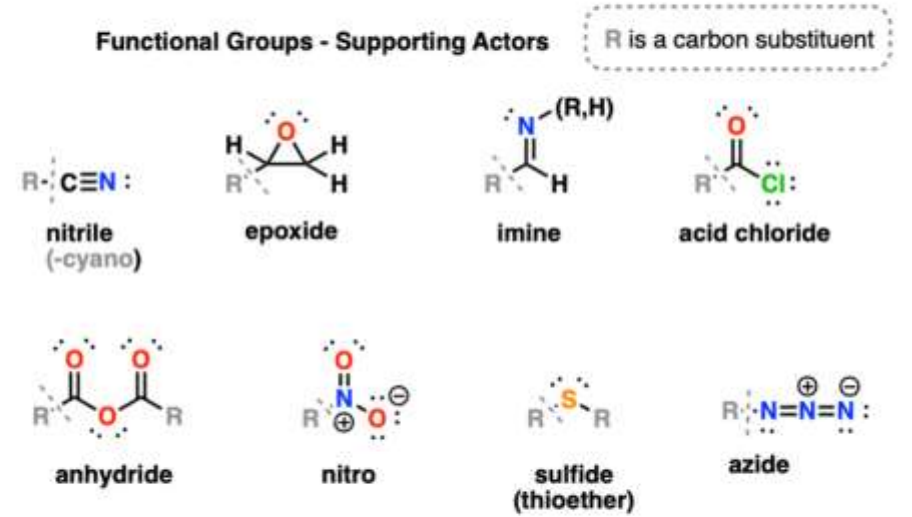
- Biocompatibility
- Bioinertness
- Biodegradability
- Durability
- Optical Properties
- Processing Requirements
- Surface Properties
- Physical Properties (strength, elastic modulus, etc.)
- Regulatory Compliance
- Economic Considerations

You don’t have to memorise those every word,
but you have to have a sense of understranding that some specific application require materials with specific properties such as….

Before you start

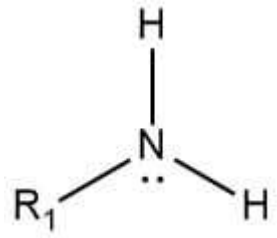
A functional group is a specific atom or group of atoms within a molecule that determines the characteristic chemical reactions and properties of that molecule. Functional groups are important in determining a compound's reactivity and the types of chemical reactions it can participate in.When considering polymers, functional groups can also change the mechanical, chemical, optical, and electrical properties of that material.



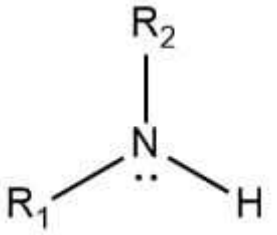


https://www.masterorganicchemistry.com/2010/10/06/functional-groups-organic-chemistry/

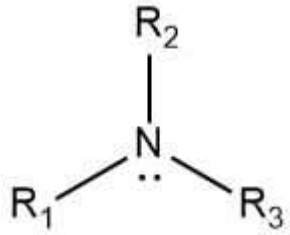
Before you start



Primary Amine



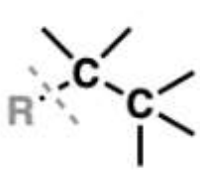
Secondary Amine



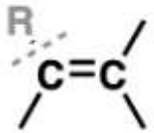
Tertiary Amine

Hydrocarbon functional groups

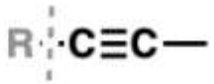
R is a carbon substituent



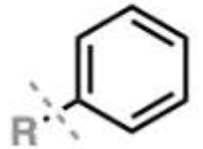
alkane
(-alkyl)



alkene
(-alkenyl)



alkyne
(-alkynyl)



benzene ring
(-phenyl)

- non-polar
- extremely weak acids

electronegativity of carbon: **2.5**
electronegativity of hydrogen: **2.2**

Properties of polymers are dependent on several factors

Chain length

Crystallinity

Molecular weight

Branching

Cross-linking

Additives

Density

Chemical structure (functional groups)

Porosity

Etc.,

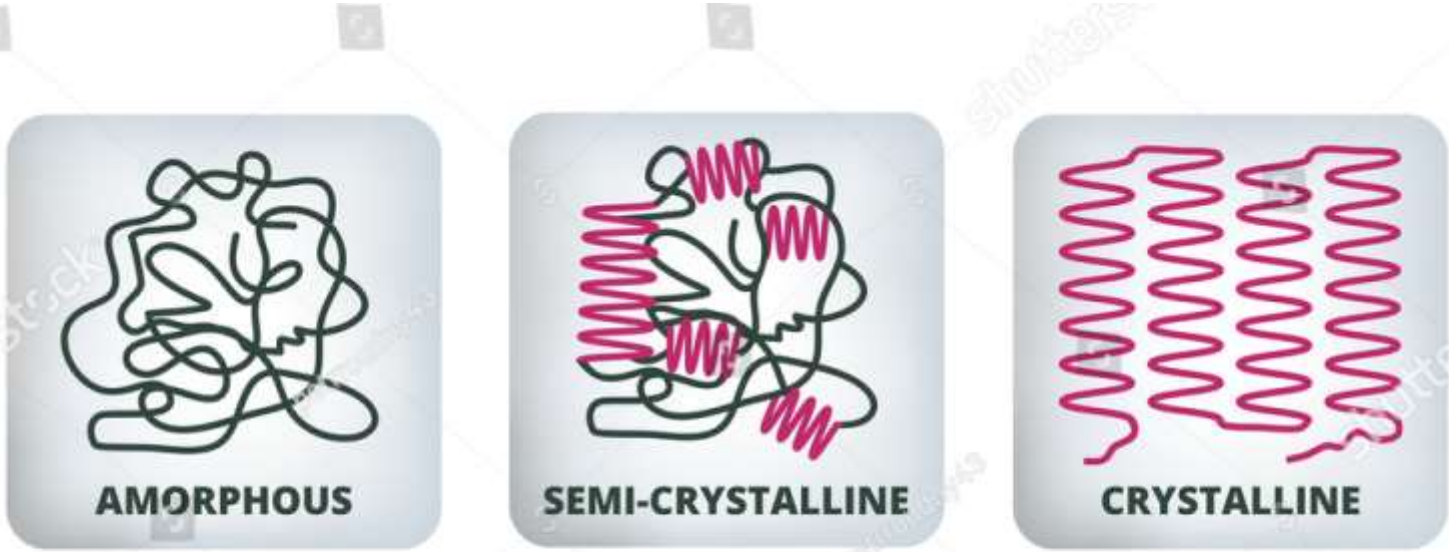
Crystallinity of polymers

Partiularly melting temperature and glass transition temperature are important aspects in the process of selection of polymers.

Glass transition temperature (T_g): Non-crystalline material is hard and brittle below T_g. However, above Tg, the material transforms into a rubbery substance (glass transition is observed in amorphous or semicrystalline polymers only not in crystalline ones!)

Melting: change of a solid into a liquid when heat is applied substance.

Determination of polymer crystallinity is important because mechanical and even optical properties of crystalline and amorphous polymers differ. A crystalline polymer tend to have higher melting point compared to amorphous counterpart.



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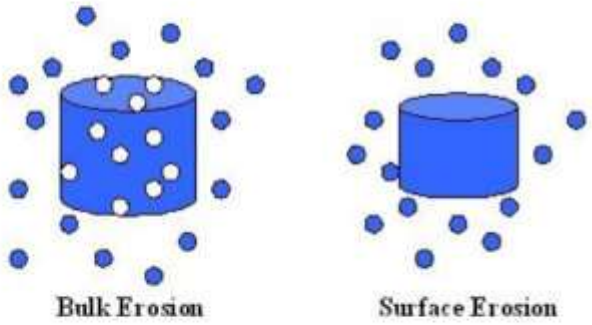
Biodegredable polymers

Depending on the application one may prefer polymers that are biodegredable or bioinert.
Polymers have had a long history of use as inert materials for implants, dental materials, sutures, contact lenses, pharmaceutical excipients, and more.
There are also biodegredable polymers.

Biobozunur Polimerler

Bozunma mekanizmaları

- Enzimatik
- Hidroliz
  - Homojen
  - Heterojen
- Yüzey Erozyonu
  (Poly(ortho)ester)
- Bulk Bozunma
  (PLA,PGA,PLGA, PCL)



Enzymatic biodegradation occurs when **microorganisms** like bacteria or fungi secrete **enzymes** that break down the polymer. Enzymes produced by the human body can also degrade polymers.

Hydrolysis is the breakdown of a polymer by **reaction with water**. It is especially common in polymers that contain hydrolyzable functional groups, such as **esters**, **amides**, or **anhydrides**.

Hydrolysis: any chemical reaction in which a molecule of water breaks one or more chemical bonds.

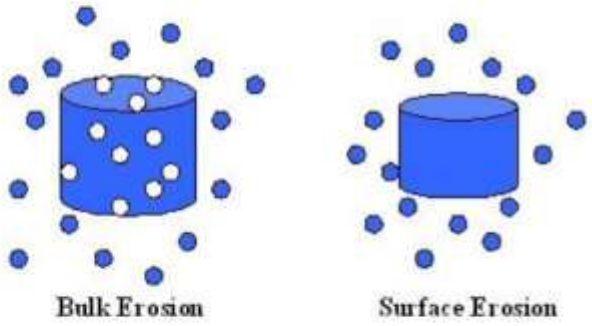
Biodegredable polymers

Surface erosion and bulk erosion describe where degradation and material loss occur

Biobozunur Polimerler

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In **surface erosion**, the degradation occurs mainly at the **surface** of the polymer, while the interior remains largely unaffected for some time.

Bulk erosion occurs throughout the entire polymer structure rather than just at the surface. The water or degrading agent penetrates the polymer, causing the material to degrade evenly in the interior as well as at the surface. Degradation occurs throughout the volume, but it is not necessarily uniform.

Biocompatibility of polymers

Depending on the application the rate of biodegredation is also important. Such as sutures are produced from biopolymers that degrade rapidly. Biodegredability can be controlled through controlling molecular structure.

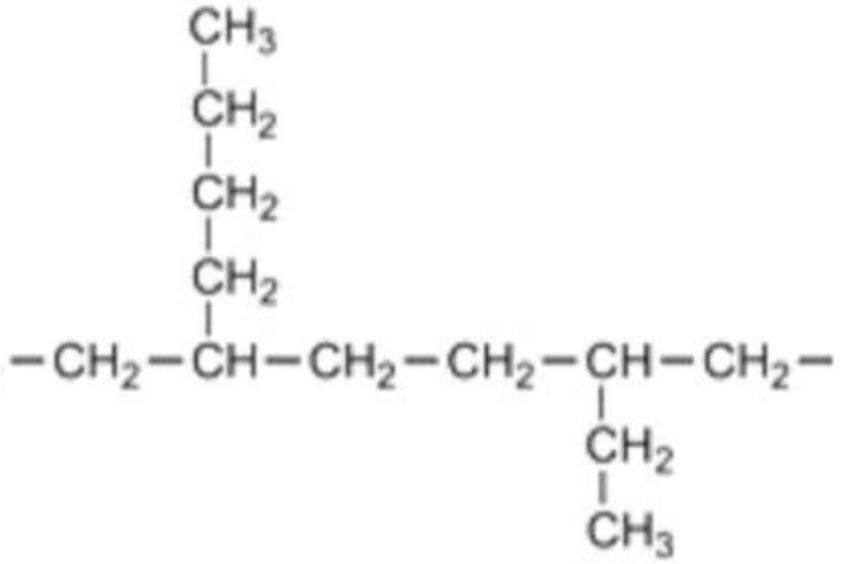
Natural biopolymers

Natural polymers themselves can be divided into two subcategories: polysaccharides and proteins. Common polysaccharides used in bone tissue engineering include alginate, chitin/chitosan, hyaluronic acid derivatives, cellulose and gelatin as well as proteins such as collagen, and silk fibroin.

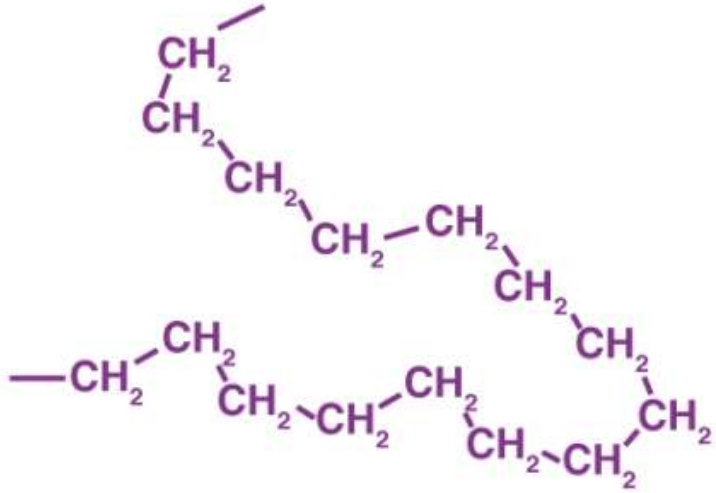
Chitosan is one of the prominent candidates and one of the most widely used natural polymers in the field of bone tissue engineering. Chitosan is biodegredable.

Polyethylene

Low-Density Polyethylene (LDPE)



High-Density Polyethylene (HDPE)

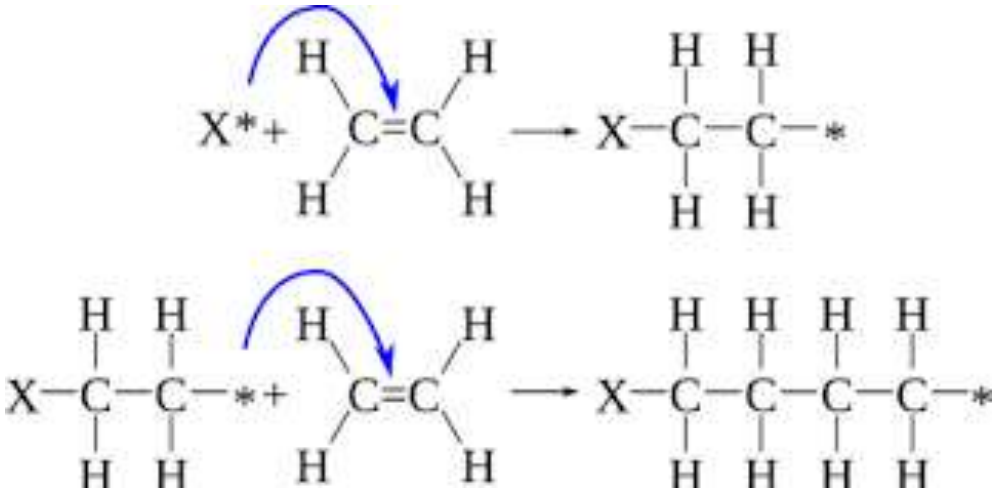


Low-Density Polyethylene (LDPE): LDPE has relatively short and branched polymer chains. It is more amorphous and has a lower density, which creates a flexible and soft material.
High-Density Polyethylene (HDPE): HDPE has longer and more linear polymer chains.

For high mechanical strength, PE has to be linear.

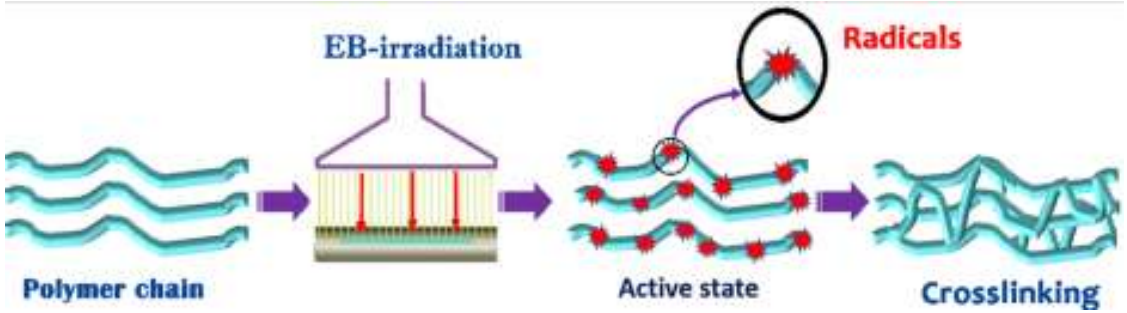
Ultra-High Molecular Weight Polyethylene

UHMWPE (Ultra-High Molecular Weight Polyethylene) has an **extremely high molecular weight**, typically in the range of **3 to 6 million g/mol**, which gives it different properties.



free radical initiator ($X^{\cdot}$)

Gamma irradiation can induce **cross-linking** between polymer chains. This means that individual polymer chains are chemically bonded together at certain points, forming a more interconnected and robust network.



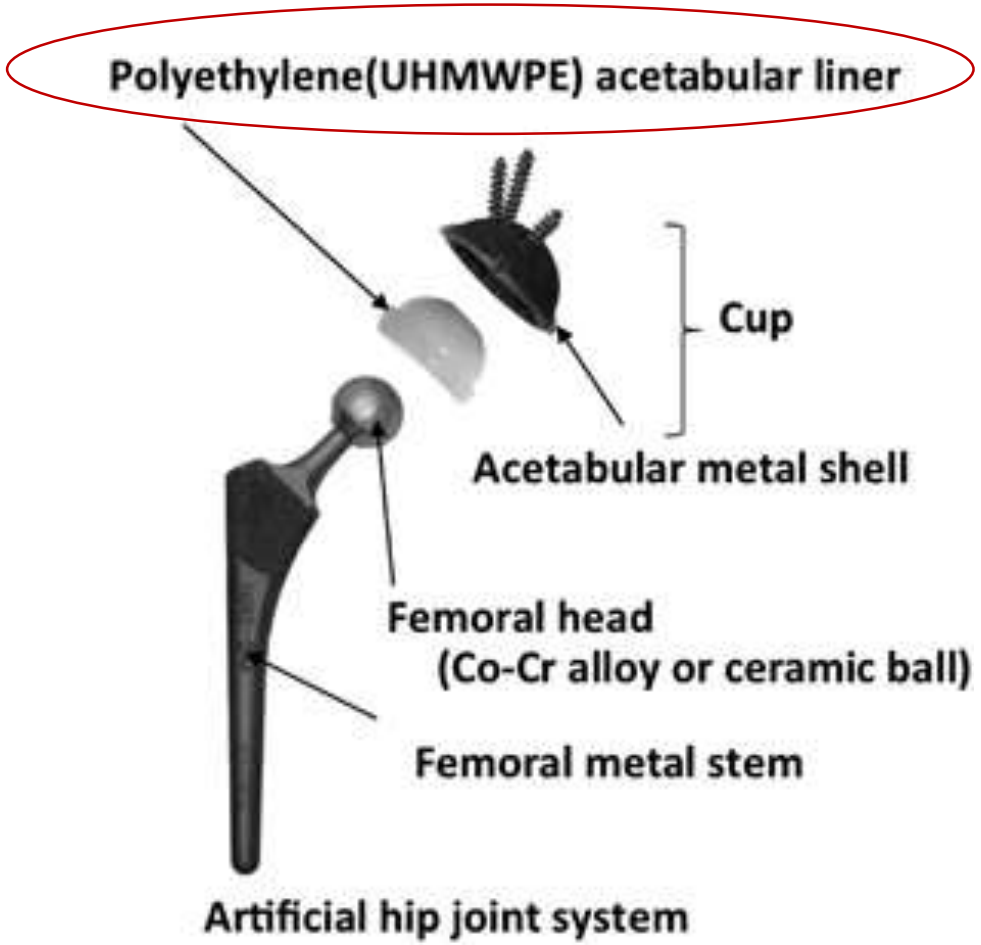
The polymer chains in UHMWPE are significantly longer than those in HDPE, which gives the material extraordinary strength and toughness.

Ultra-High Molecular Weight Polyethylene

Due to its very high molecular weight, UHMWPE has **exceptional abrasion resistance**, **impact strength**, and **low friction**.

Ultra-high molecular weight polyethylene (UHMWPE) offer excellent mechanical properties similar to natural bone, high impact, wear resistance, low friction coefficient, and biocompatibility for prostheses, they lack the ability to directly bond with bone tissue. For this reason, UHMWPE is widely used to develop knee, hip, and shoulder prostheses.

Attempts to make UHMWPE bioactive have been made by adding hydroxyapatite (HA) to it.

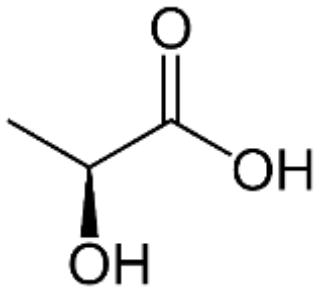


https://doi.org/10.1038/pj.2015.45

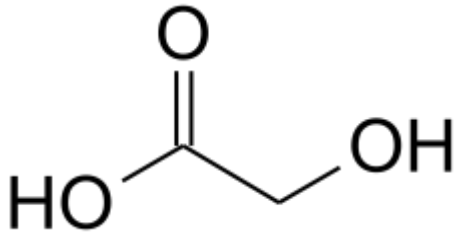
PLA, PLGA, and PGA

PLA: **Hydrophobic**, meaning it repels water, Biodegrades through hydrolysis

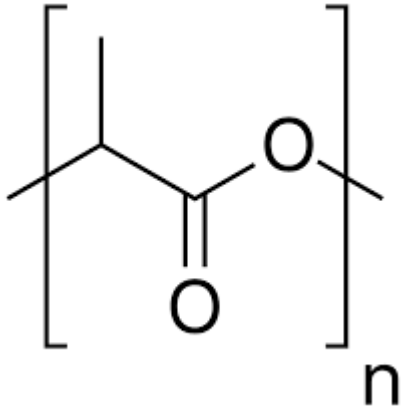
PGA: **Hydrophilic**, meaning it absorbs water, which accelerates its degradation, PGA degrades much faster than PLA.

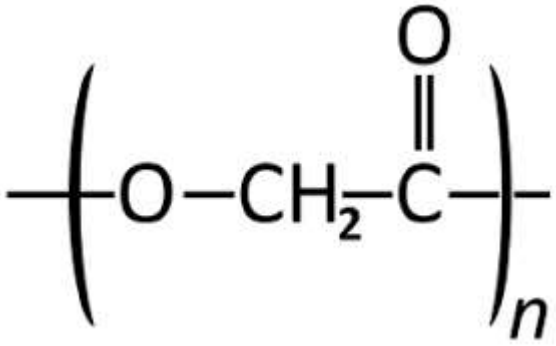


Lactic aci
d



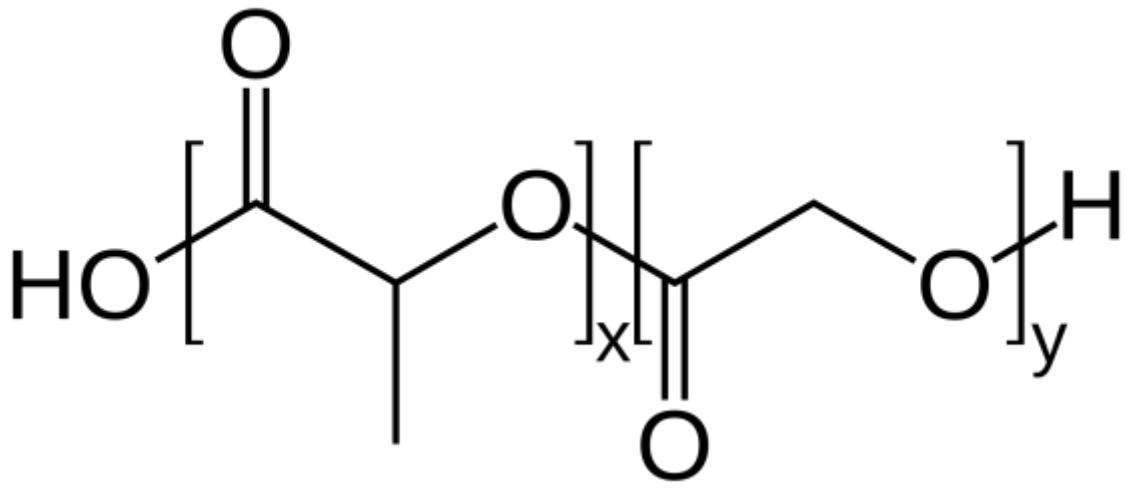
Glycolic acid





PLA, PLGA, and PGA

poly(lactic-*co*-glycolic) acid is a copolymer which is used in many medical devices, such as, grafts, sutures, implants, prosthetic devices, surgical sealant films, micro and nanoparticles.



Depending on the ratio of lactide to glycolide used for the polymerization, different forms of PLGA can be obtained.

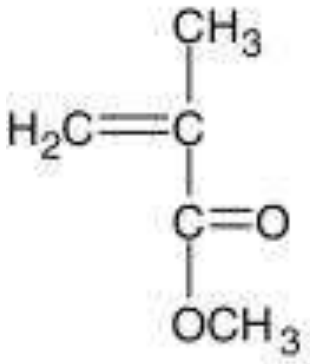
PLGA degrades by hydrolysis of its ester linkages in the presence of water.

PLGA has adjustable biodegradability

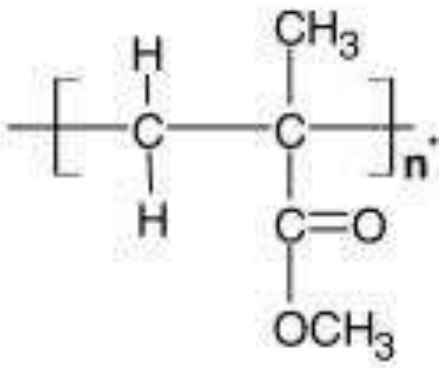
Polymerthylmetacrylate

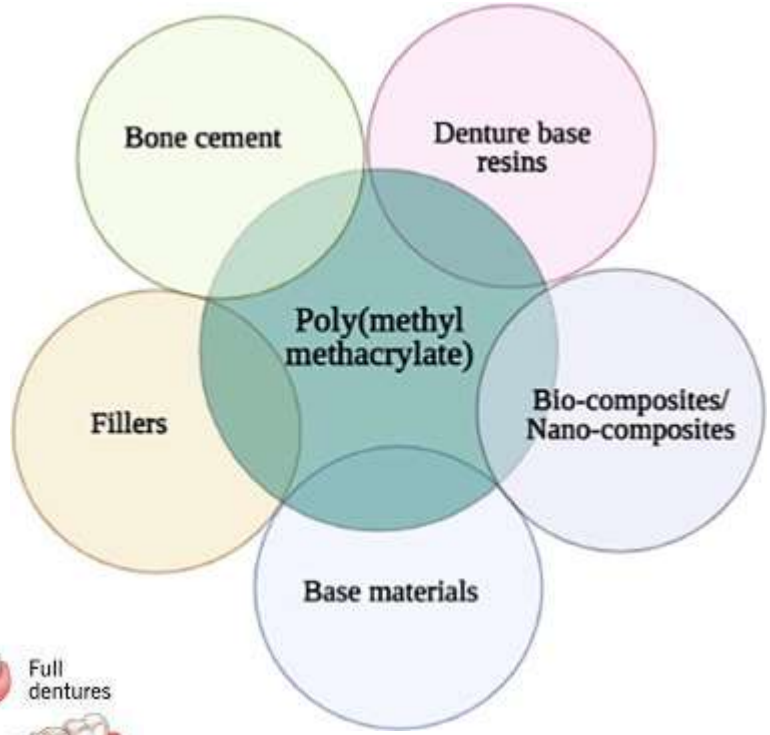
Prothesis applications require fixation to bone that relies on using poly (methyl methacrylate) (PMMA) as a bone cement, which provides initial fixation but hinders osseointegration, the crucial process of bone growth onto the implant

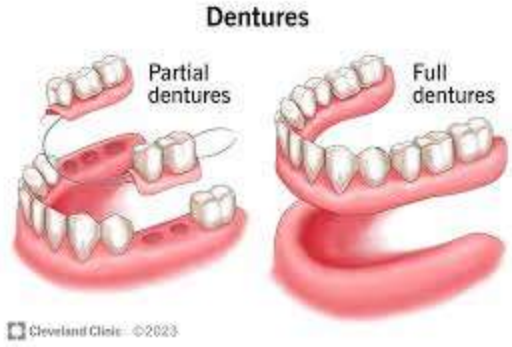
MMA Monomer



PMMA Polymer

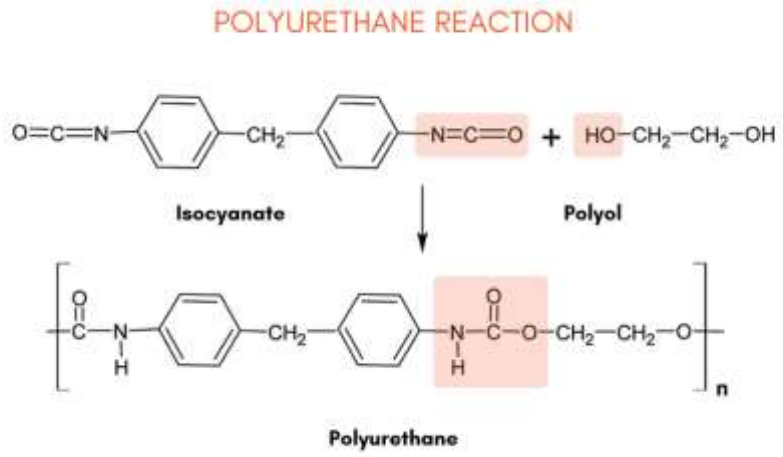


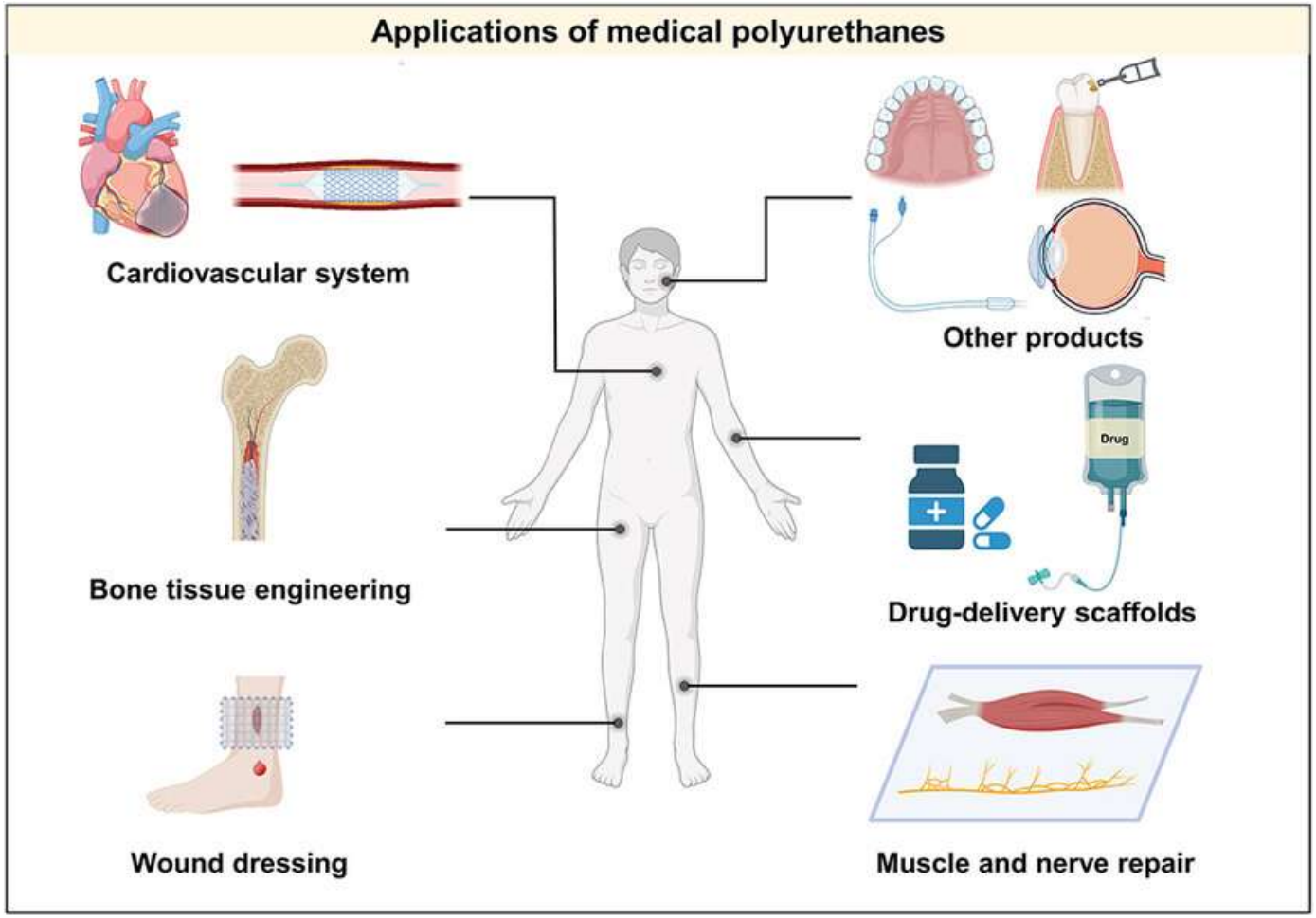




Polyurethane

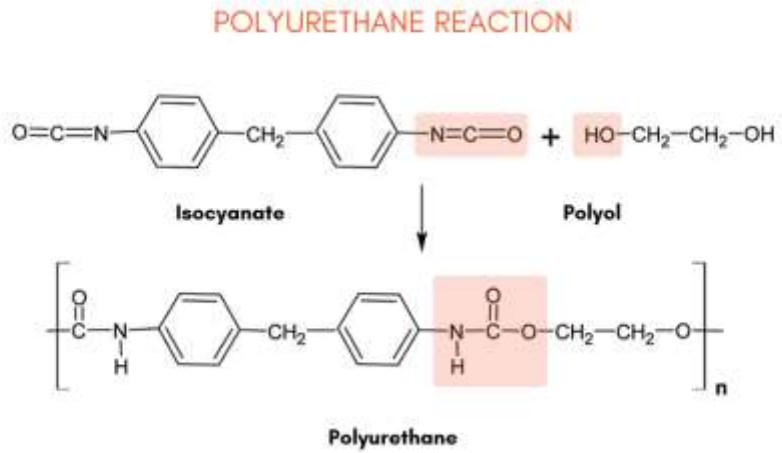
Polyurethane (PU) is mainly composed of diisocyanate as hard chain segments and oligomeric polyols as soft chain segments.





Polyurethane

Polyurethane (PU) is mainly composed of diisocyanate as hard chain segments and oligomeric polyols as soft chain segments.



Applications:
Skin sensors
Wound dressings
Vascular grafts
Catheter tubing

Vascular grafting is a surgical procedure used to repair or replace damaged blood vessels in the body. This involves the use of a graft, which is a piece of synthetic or natural material that is implanted into one to create a new blood vessel.

A dressing or compress is a piece of material such as a pad applied to a wound to promote healing and protect the wound from further harm.

A catheter is a tube that is inserted into your bladder, allowing your urine to drain freely.

Polyurethanes

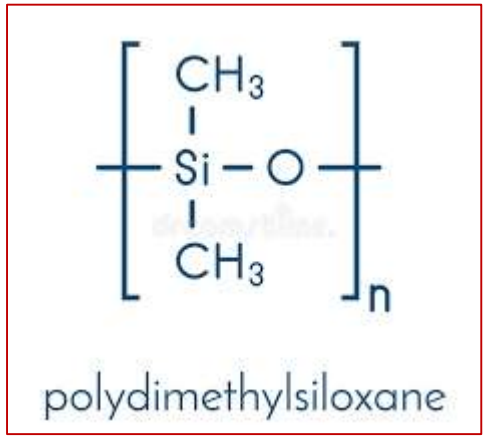
The physical properties of **polyurethane** (PU) can vary significantly based on the **synthesis conditions**, particularly the **degree of cross-linking**.

PU is biocompatible and have low water permeability, but disadvantages is isocyanonides have toxicity issues.

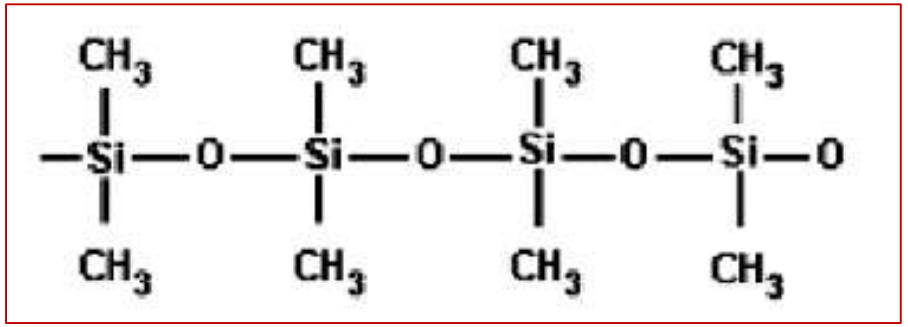
Silicones

Polymer backbone is composed of alternating silicon and oxygen atoms.

Siloxane is the basic repating unit



Silicones are water repellent, heat stable, and very resistant to chemical attack



Poly(dimethylsiloxane) (PDMS)

Advantages

Disadvantages

Relatively biologically inert, but can provoke a foreign body response.

Advantages
- Very low \(T_g\) (~−120 °C)
- High flexibility
- Good thermal stability
- Good biostability
- High oxygen permeability
- Electrically insulating

- Low stiffness and limited load-bearing capability
- Poor abrasion and tear resistance in some formulations
- Hydrophobic surface
- Nonspecific protein adsorption
- Potential fibrous encapsulation

In both blood contacting and ophthalmic applications, the ability to prevent the nonspecific adsorption of proteins is essential for proper function. So silicones have a problem for that type of applications.

Silicones

Catheters

Drainage tubes

Contact lenses

Medical adhesives

Coating for pacemakers





Hydrogels

A hydrogel is **a crosslinked hydrophilic polymer that does not dissolve in water**. They are highly absorbent (capable of absorbing large amounts of fluid) yet maintain well defined structures. These properties underpin several applications, especially in the biomedical area. Many hydrogels are synthetic, but some are derived from nature.

Hydrogel products constitute a group of polymeric materials, the hydrophilic structure of which **renders them capable of holding large amounts of water** in their three-dimensional networks.

They have very high water absorption capacity (90%wate



They can have controllable degradability.

Hydrogels

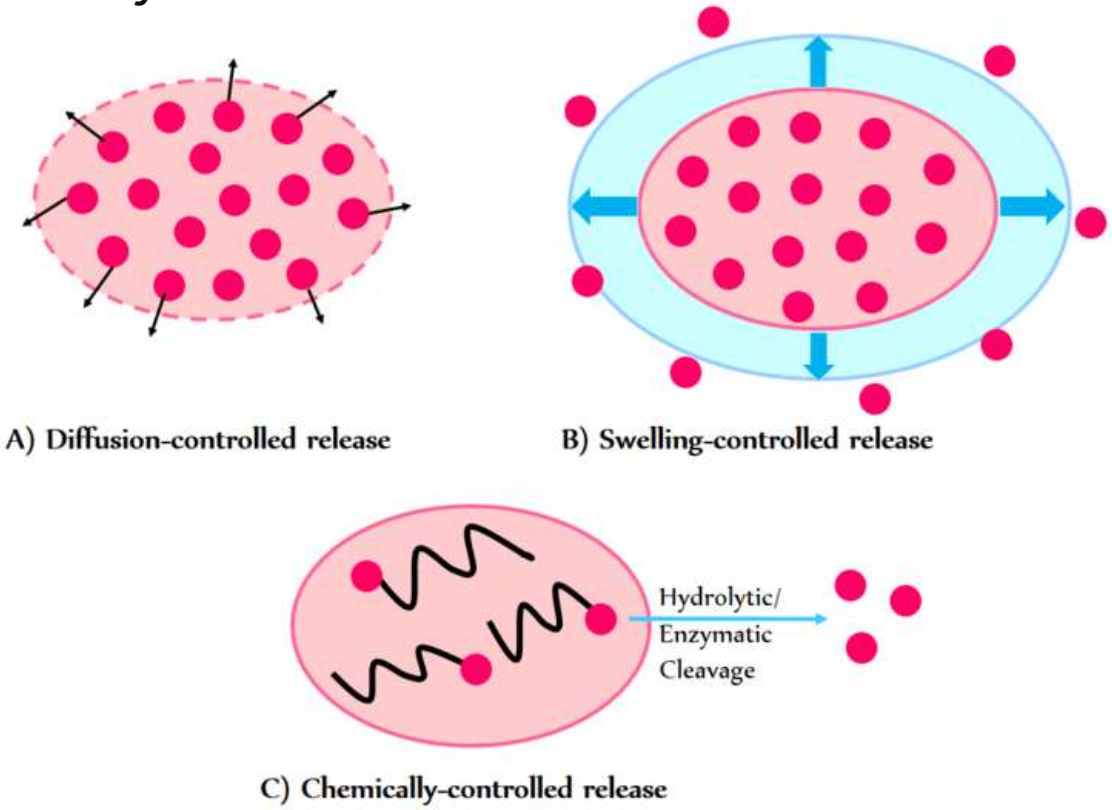
Uses:

drug delivery,
biosensors,
wound dressings,
bioelectronic devices,
contact lenses, etc.

The mechanism of drug delivery in hydrogels is often based on their ability to swell or contract in response to external triggers like pH, temperature, or other stimuli, which in turn controls the release of the drugs they contain.

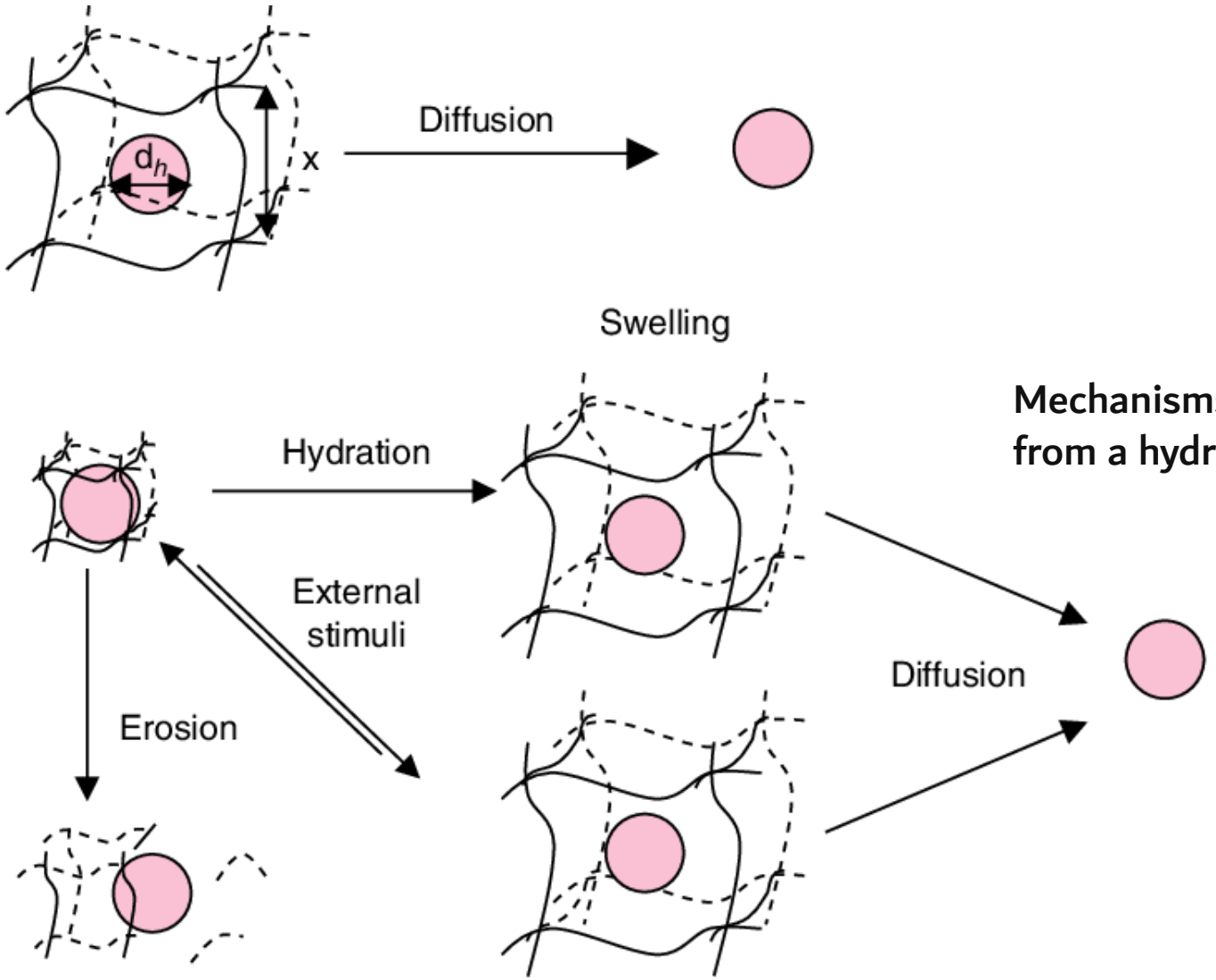
They are capability to easily encapsulate hydrophilic drugs.

Schematic view of drug release mechanisms from hydrogels; A) diffusion-controlled release, B) swelling-controlled release, and C) chemically-controlled release.



A drug delivery systems can control how the drugs are available to cells and tissues over time and in space.

Hydrogels

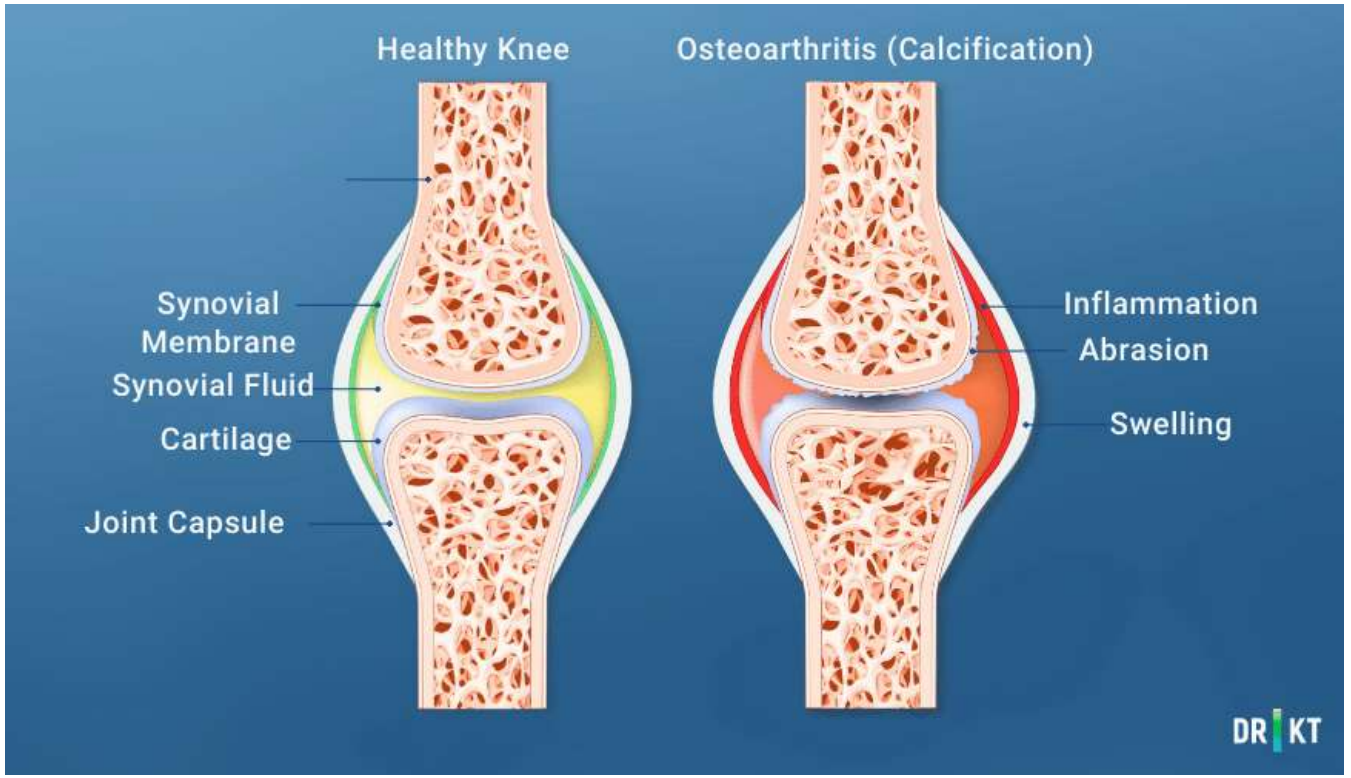


Mechanisms potentially involved in the drug release process from a hydrogel-based formulation

10.2165/00137696-200301020-00001

Hydrogels

Hydrogel Treatment for Knee Osteoarthritis



Hydrogel injection can help reduce synovial inflammation by creating a stable and long-lasting layer covering the surface of the synovial membrane.

Hydrogel acts as a scaffold. In other words, it creates a kind of cushion inside the joint

Hydrogels: molecular architecture

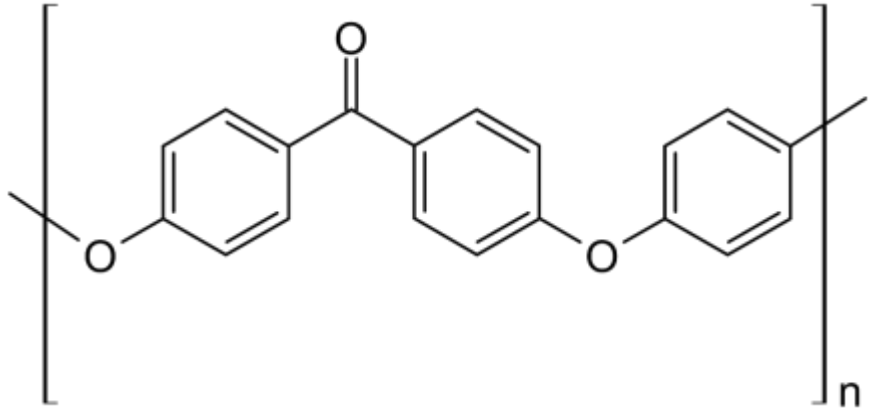
They are synthesized by a wide range of polymerization methods that allow for the introduction of linear and branched units with specific molecular characteristics.

Monomer chemical name	Monomer abbreviation	Properties
Hydroxyethyl methacrylate	HEMA	Hydrophilic, hydroxyl functional
N-vinyl-2-pyrrolidone	NVP	Hydrophilic
Acrylic acid	AA	Hydrophilic; acid-containing
Ethylene glycol	EG	Hydrophilic; H-bonding site

poly(ethylene glycol) (PEG)

poly(methacrylic acid)

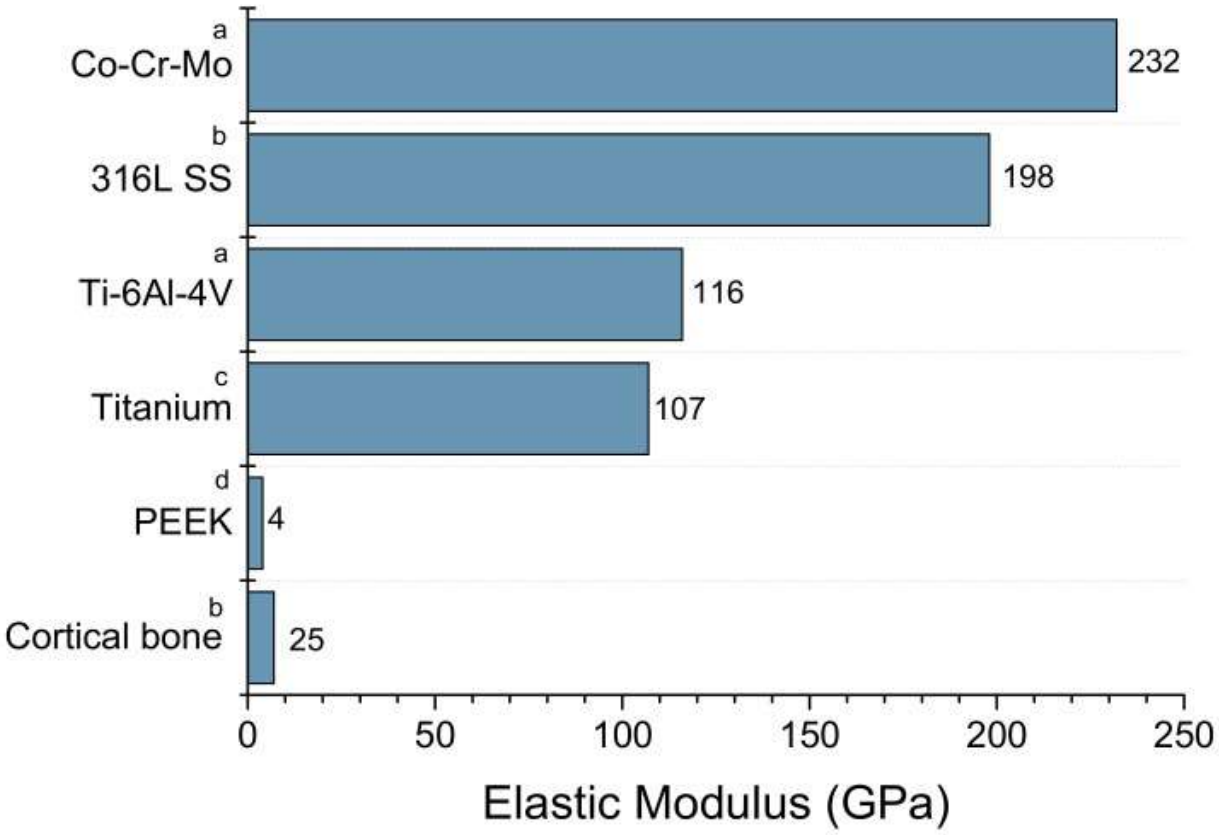
Poly (Ether-Ether-Ketone)



Aromatic Rings: PEEK contains aromatic rings in its polymer chain, which provide strength and stiffness. These rings are more stable and less prone to deformation than aliphatic structures found in other polymers.

Crystallinity: PEEK has a crystalline structure, meaning its polymer chains are arranged in an ordered pattern. This arrangement contributes to its strength and stiffness.

Polyether ether ketone (PEEK)



Elastic modulus of bone, PEEK, and metals usually used in implants

Dental implants
Dental implant abutments
Orthodontic wires
Prothesis



Need improvement in properties



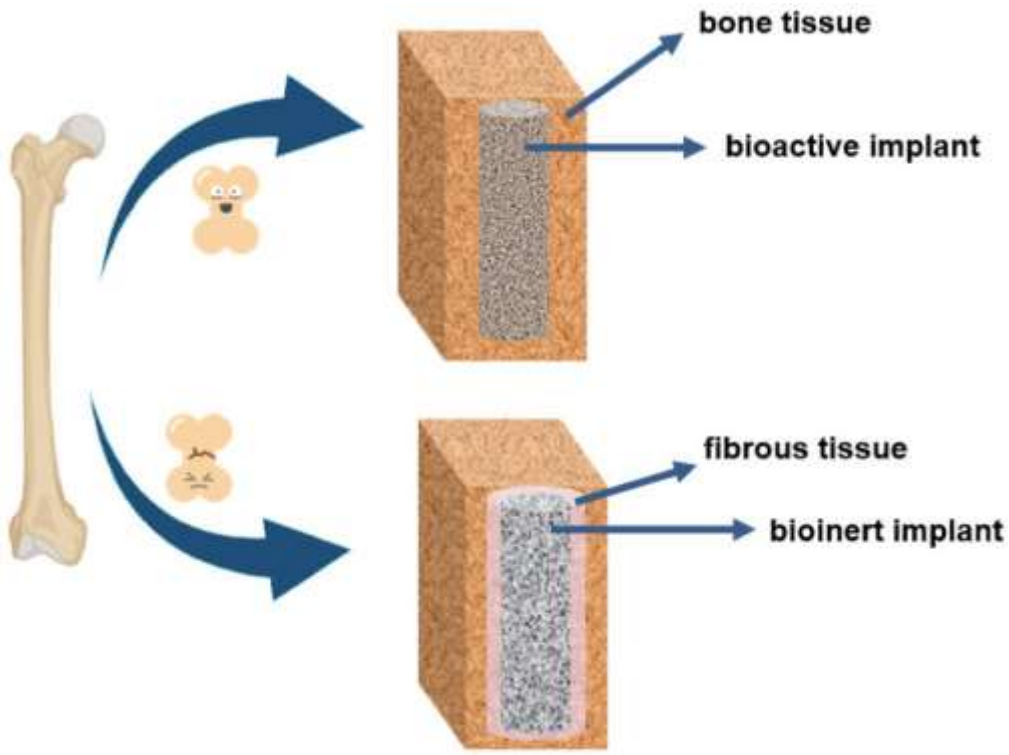
Nano sized coated PEEK implant with hydroxyapatite particles

PEEK

PEEK has hydrophobicity and lack of bioactivity.

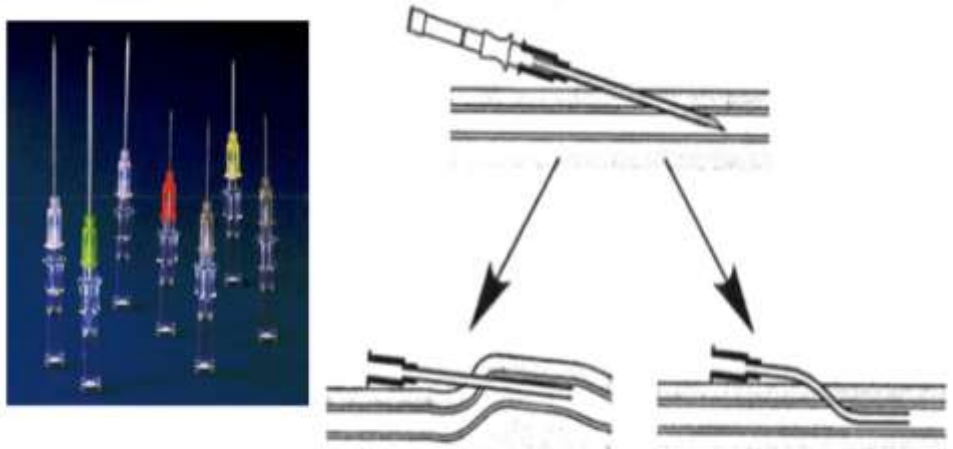
Surface modification: surface modification technique aims to alter the surface characteristics of PEEK without affecting its bulk properties.

Bulk modification: compounding it with nanoparticles such as hydroxyapatite.

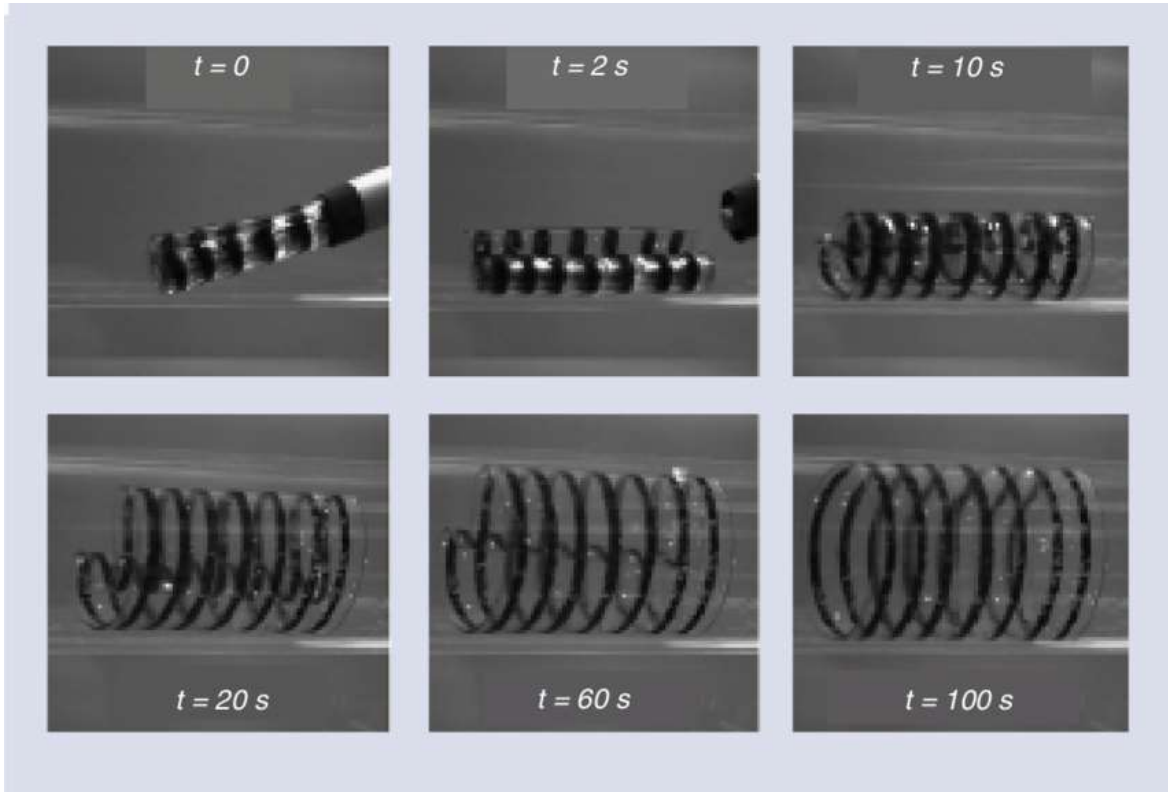


Shape memory polymers

Polymers that can recover deformation in response to a stimulus



Elastikleşen (yumuşayan) iğne



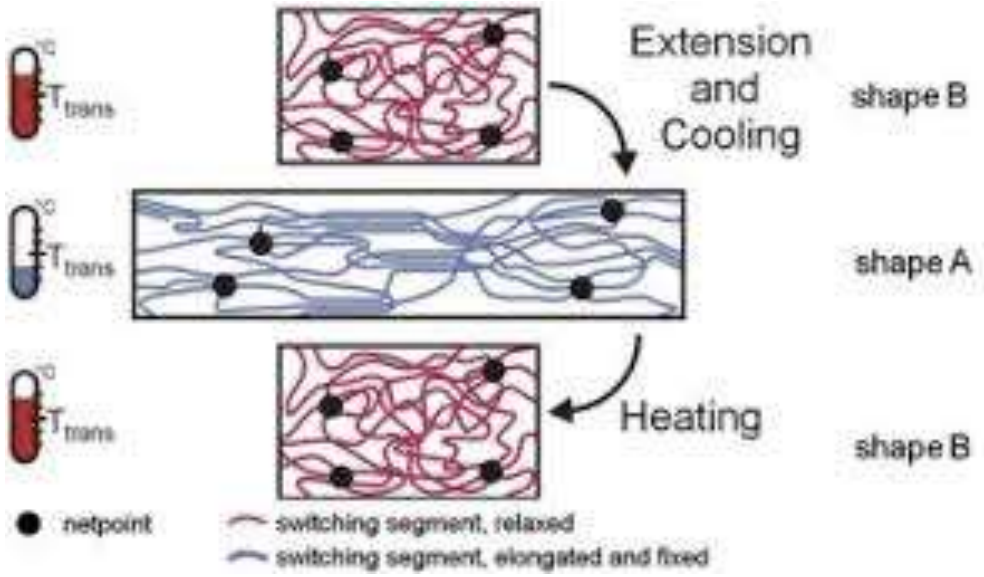
self-expandable stents

Polylactic acid is a SMP

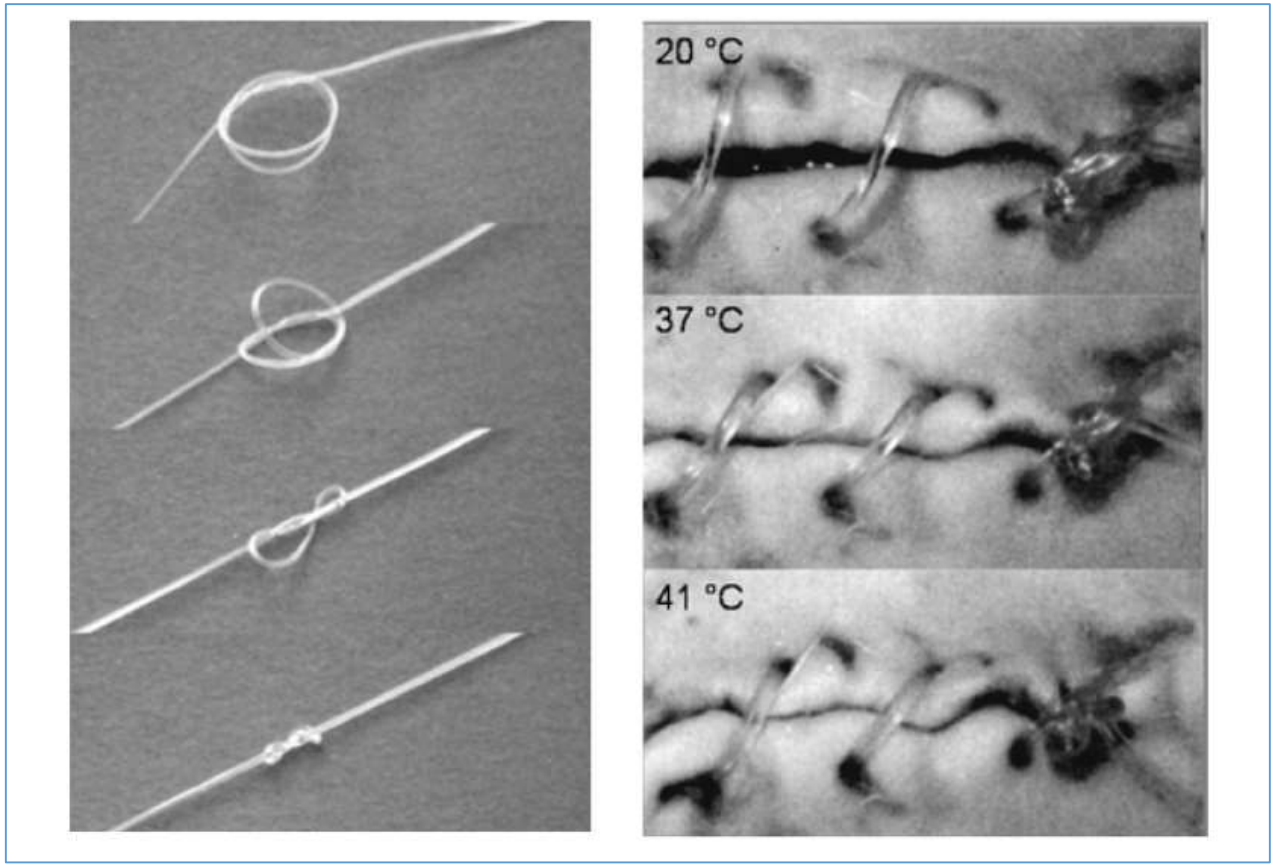
Shape memory polymers

The shape memory mechanism of polymers (SMPs) involves the ability of a polymer to "remember" its original shape and return to it when exposed to an external stimulus, such as heat.

thermal shape memory polymer



Shape memory cycle:
heating → deforming → cooling → reheating



Suture becomes tightened with body temperature.

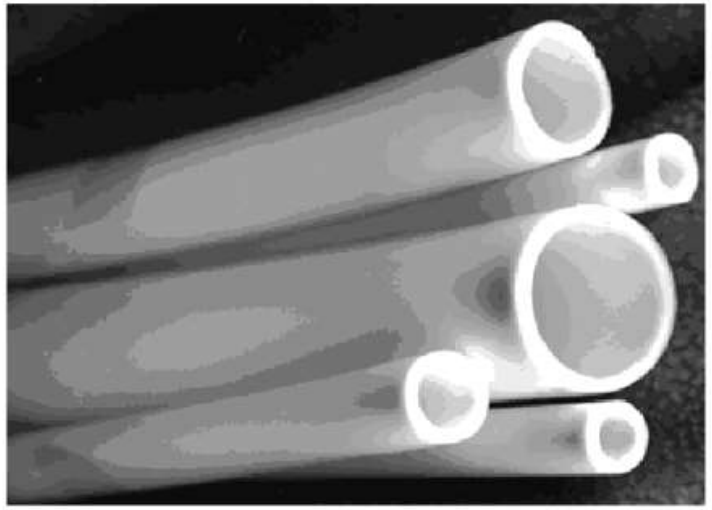
Teflon

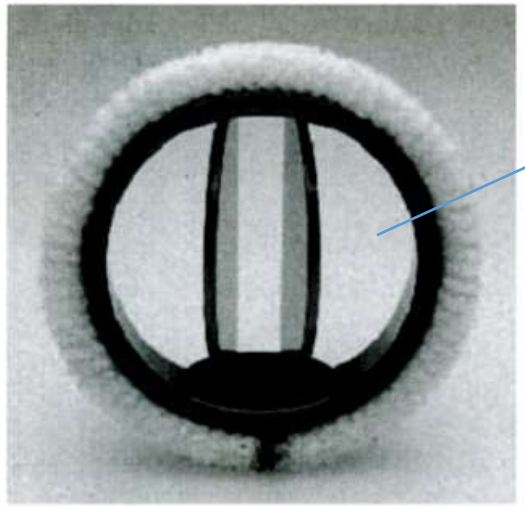
PTFE, (Teflon)

Isısal ve kimyasal açıdan kararlı ancak işlenmesi zordur. Çok hidrofobik ve mükemmel kayganlığa sahiptir.

Damar protezlerinde,

Kalp kapakçıklarında kullanılır.



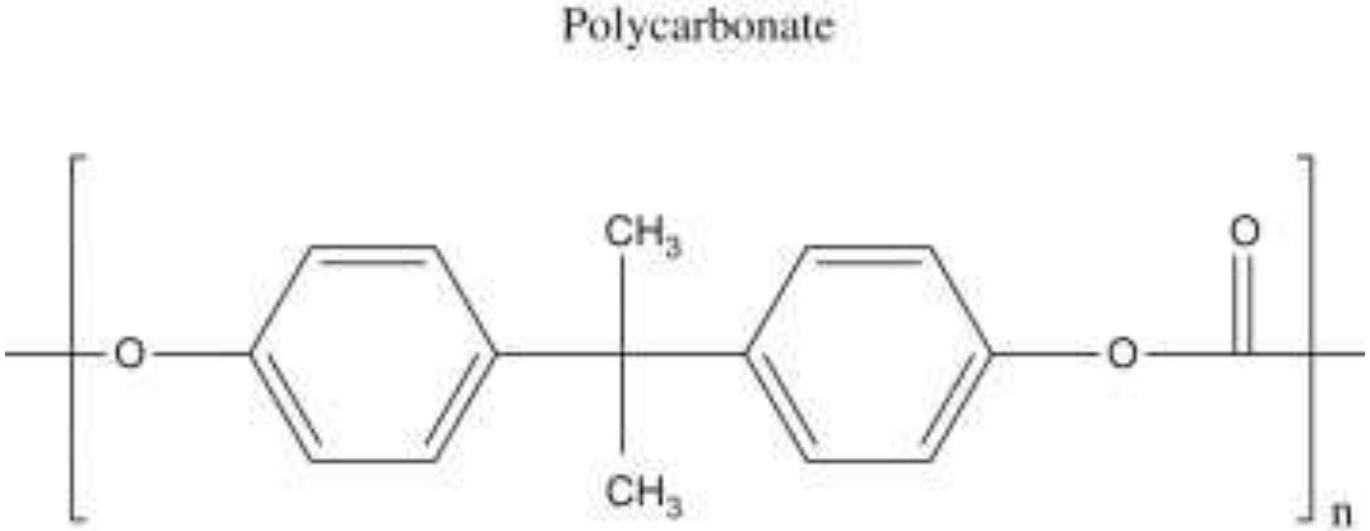


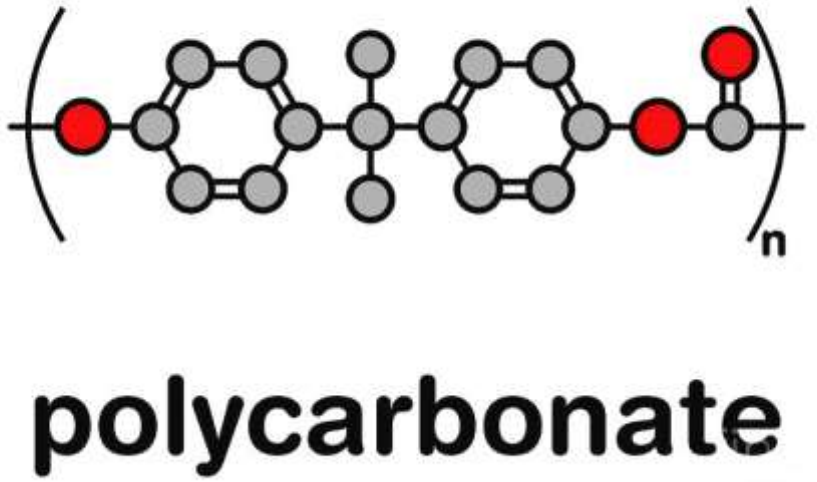
A mechanical hart valve

PTFE is needed to ensure a medical device does not accumulate any bacteria during procedures but also has high non-stick properties as well.

Polycarbonates

Applications: Dialysis components, oxygenators, syringe bodies





Bibliography

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https://doi.org/10.3390/gels10070471

10.30476/tips.2019.81604.1002

Additional: Bone grafts

Generally, bone tissue has a substantial innate capacity to self-healing from defects. There are situations related to orthopaedics and oral and maxillofacial surgery that require an extensive level of bone regeneration that can exceed the intrinsic self-healing ability of bone and, as a result, causes bone repair to be delayed and impaired; these are known as critical bone defects.

Adjunctive treatment for difficult fractures and bone defects using porous bone substitutes (scaffolds) is required so that bone and vascular cells may colonize the scaffold and new bone tissue with similar functions may develop. Autologous bone grafting is widely regarded as the preferred method for reconstructing small bone defects being most biocompatible.

Biomaterials that mimic the structure of the natural extracellular matrix (ECM) have the ability to create a microenvironment similar to bone.

Polimer İmplantlar

- ✓ Kardiyovasküler alanda (PU, PTFE, Silikon Kauçuğu)
- ✓ İlaç salım sistemlerinde (PLG, PLA)
- ✓ Doku mühendisliğinde iskelet malzemesi (PLA, PLG)
- ✓ Kalp kapakçıkları (PTFE)
- ✓ Yapay kalpler (PU)
- ✓ Lensler (PMMA, PC)
- ✓ Ortopedik implant (UHMW-PE, PMMA)

Biyobozunur Polimerler

Doğal

- ✓ Fibrin
- ✓ Kolajen
- ✓ Kitosan
- ✓ Jelatin
- ✓ Hyaluronan

Sentetik

- ✓ PLA, PGA, PLGA, PCL, Polyorthoesters
- ✓ Poly(dioxanone)
- ✓ Poly(anhydrides)
- ✓ Poly(trimethylene carbonate)
- ✓ Poly(phosphazenes)

Additional: Contact lenses

Contact lenses are primarily made from specialized **materials** that offer the required **oxygen permeability**, **biocompatibility**, and **comfort** for long-term contact with the eye.

Polymers:

Polymethyl methacrylate (PMMA)

Hydrogels

Silicone Hydrogels

Fluorosilicone acrylate and **silicone acrylate**

Bibliography

Li and Mooney, Designing hydrogels for controlled drug delivery, Nature Reviews Materials (2016).